

Breastfeeding Practices And Its Associations With Nutritional Intake Adequacy, Blood Pressure And Body Mass Index In Postpartum Mothers: A Cross-sectional Study

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Introduction

Nutritional focus often shifts entirely to the newborn, causing up to 57% of postpartum mothers to fail to meet basic energy requirements (Carretero-Krug *et al.*, 2024). Unbalanced or inadequate postpartum diet due to traditional confinement diet can triggers cardiovascular and metabolic disorders among postpartum mothers who are breastfeeding (Berezowsky and Berger, 2021). This study aimed to assess postpartum mothers’ dietary and nutrient intake adequacy according to the Malaysian Recommended Nutrient Intake (RNI) 2017 and to examine its associations with blood pressure and body mass index (BMI).

Methodology

A cross-sectional study was conducted at Greentown Health Clinic, Perak, Malaysia from June 2024 - January 2025 among postpartum mothers who are breastfeeding for infants aged below 2 years old.

Data collection tools:

- Self-designed questionnaire assessed on sociodemographic, obstetric & medical, breastfeeding practices
- 24-Hour dietary recall record (Faber *et al.*, 2016)

Blood pressure measurement and Body mass index:

Blood pressure data was measure using automatic BP monitor (Omron), and BMI was measured using Omron Karada scan.

Ethical approval:

- UTAR Scientific Ethical & Research Committee (SERC56B-802025).
- National Medical Research Registry (NMRR)(ID-24-02012_WYI).

Data Analysis:



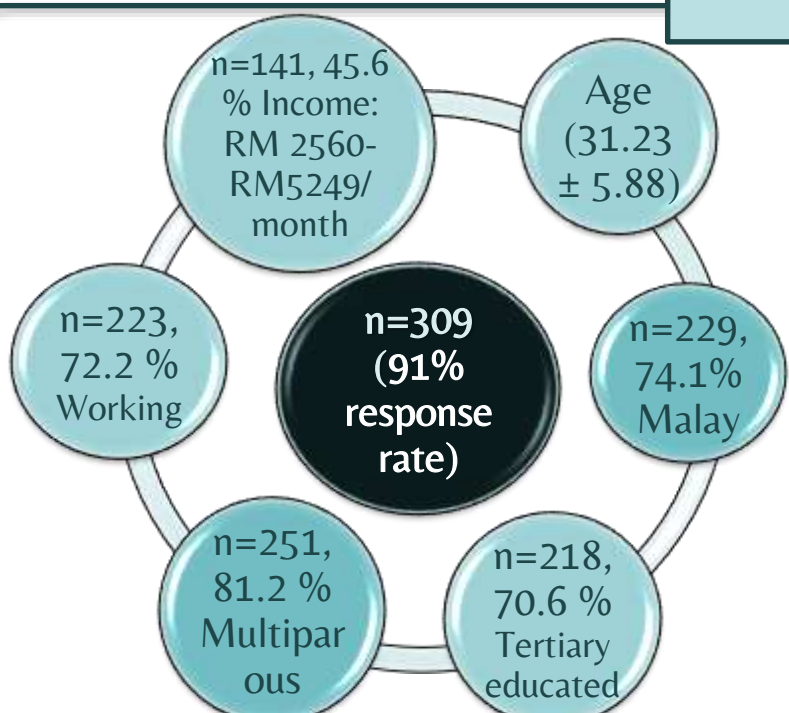
- ✓ Chi-Square test
- ✓ Wilcoxon Rank test
- ✓ Mann-Whitney U test

Main Reference:

- Recommended Nutrient Intake (RNI) 2017 (Ministry of Health Malaysia, 2017)



Results



Variables	Mean	SD	Total	
			n	%
Blood Pressure (BP)				
Systolic	114.68	13.84		
Normal (< 120 mmHg)			211	68.3
Elevated & Hypertension (≥120 mmHg)			98	31.7
Diastolic	73.80	7.80		
Normal (≤ 80 mmHg)			249	80.6
Hypertension (> 80 mmHg)			60	19.4
Combined BP (mmHg)				
Normal			164	53.1
Elevated			53	17.2
Hypertension			92	29.8
BMI (kg/m²)	26.59	4.08		
Underweight & normal (< 25.0 kg/m²)			110	35.6
Overweight (25.0-29.5 kg/m²)			134	43.4
Obese (> 30.0 kg/m²)			65	21.0

Postpartum mothers’ blood pressure (< 120 mmHg) (n= 211, 68.3%) , and BMI (< 25.0 kg/m²) (n=110, 35.6%). Mothers of infants > 6 months had significantly different systolic BP (p < 0.001) and diastolic BP (p=0.043) of postpartum mothers with infants ≤ 6 months while BMI did not show statistically significant variation (p > 0.05).

Nutrients	Mean intake	% meeting RNI	Wilcoxon Z
Caffeine (mg)	25.32	-	-
Calcium (mg)	411.02	1.62	-15.157
Copper (mg)	7.91	25.89	-8.931
Fluoride (µg)	1065.96	-	-
Folate (DFE, µg)	396.95	21.68	-10.326
Iron (mg)	18.83	60.52	-4.478
Magnesium (mg)	182.32	7.12	-14.941
Manganese (mg)	2.94	18.77	-10.680
Niacin (mg)	18.83	54.05	3.160
Phosphorus (mg)	1200.49	77.67	11.716
Potassium (mg)	1689.34	0	-15.236
Riboflavin (µg)	8.85	33.33	-6.083
Selenium (µg)	62.91	79.61	-12.162
Sodium (mg)	2470.38	77.35	11.159
Thiamine (µg)	1.45	30.74	-7.119
Vitamin A (RE, µg)	756.19	33.66	-4.780
Vitamin C (mg)	52.42	17.15	-10.694
Vitamin K (µg)	26.01	8.09	-14.06
Zinc (mg)	7.09	17.80	-11.170
Cholesterol	286.45	-	-
Total calories (kcal)	1659.93	26.86	-8.816
Carbohydrate (g)	220.50	-	-
Fat (g)	52.28	-	-
Protein (g)	76.12	-	-3.433
MAR*	0.66	-	-

Majority of postpartum mothers practiced exclusive breastfeeding (n=179, 57.9%). Postpartum mothers’ macro and micronutrients intake were met the Malaysian RNI for adequate carbohydrates (n=172, 55.7%), protein (n=181, 58.6%), fat (n=60, 19.4%), calcium (n=5, 1.62%), magnesium (n=22, 7.12%), potassium (n=0, 0%), vitamin K (n=25, 8.09%), vitamin C (n=53, 17.15%), zinc (n=55, 17.8%), manganese (n=58, 18.77%), folate (n=67, 21.68%), copper (n=80, 25.89%), and total calories (n=83, 26.86%). There is no significant association between breastfeeding practices and overall energy or macronutrient adequacy (p > 0.05). *MAR (mean = 0.66; SD = 0.12; IQR: 0.58 - 0.75) indicate the mothers inadequate consumption of micronutrients.

Discussions

- This demographic pattern is broadly consistent with clinic-based maternal-health research in Malaysia, particularly in settings with a substantial Malay population (Shohaimi *et al.*, 2022).
- This study finding suggests relatively favourable breastfeeding practices among the mothers in the present study (Ministry of Health Malaysia, 2016).
- Despite the relatively high prevalence of exclusive breastfeeding, substantial dietary inadequacies were observed.
- The dietary inadequacies observed in this study are particularly important during the postpartum period, no significant association was observed between breastfeeding practices and overall energy or macronutrient adequacy (p > 0.05) Agustina *et al.*, 2023; Norbaya *et al.*, 2025).
- The mother’s blood pressure may vary across the postpartum period and warrants continued monitoring beyond the early postpartum months.
- Postpartum nutritional counselling should therefore complement breastfeeding support.

Conclusion

Substantial dietary and nutrient inadequacies were observed among postpartum mothers. Systolic and diastolic blood pressure differed significantly according to infant age, while BMI did not show significant variation. Breastfeeding practices were not significantly associated with overall energy or macronutrient adequacy. These findings highlight the need for postpartum care that integrates breastfeeding support with dietary assessment, nutrition counselling, and routine monitoring of maternal blood pressure and anthropometric status. Longitudinal studies are recommended to clarify temporal relationships between variables.

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Acknowledgement & Declaration of conflict

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